$$(A\times B)\cup(A\times C) = \{(2,0),(2,1),(3,0),(3,1)\}\cup\{(2,1),(2,2),(3,1),(3,2)\}$$
$$= \{(2,0),(2,1),(2,2),(3,0),(3,1),(3,2)\} \quad ...(2)$$

From (1) and (2), $A\times(B\cup C) = (A\times B)\cup(A\times C)$ is verified.

(ii) $A\times(B\cap C) = (A\times B)\cap(A\times C)$

$$B\cap C = \{0,1\}\cap\{1,2\} = \{1\}$$
$$A\times(B\cap C) = \{2,3\}\times\{1\} = \{(2,1),(3,1)\} \quad ... (3)$$
$$A\times B = \{2,3\}\times\{0,1\} = \{(2,0),(2,1),(3,0),(3,1)\}$$
$$A\times C = \{2,3\}\times\{1,2\} = \{(2,1),(2,2),(3,1),(3,2)\}$$
$$(A\times B)\cap(A\times C) = \{(2,0),(2,1),(3,0),(3,1)\}\cap\{(2,1),(2,2),(3,1),(3,2)\}$$
$$= \{(2,1),(3,1)\} \quad ... (4)$$

From (3) and (4), $A\times(B\cap C) = (A\times B)\cap(A\times C)$ is verified.

Note

The above two verified properties are called distributive property of cartesian product over union and intersection respectively. In fact, for any three sets A, B, C we have
(i) $A\times(B\cup C) = (A\times B)\cup(A\times C)$ (ii) $A\times(B\cap C) = (A\times B)\cap(A\times C)$.

1.3.1 Cartesian Product of three Sets

If A, B, C are three non-empty sets then the cartesian product of three sets is the set of all possible ordered triplets given by

$A\times B\times C = \{(a,b,c)$ for all $a\in A, b\in B, c\in C\}$

Illustration for Geometrical understanding of cartesian product of two and three sets

Let $A = \{0,1\}$, $B = \{0,1\}$, $C = \{0,1\}$

$A\times B = \{0,1\}\times\{0,1\} = \{(0,0),(0,1),(1,0),(1,1)\}$

Representing $A\times B$ in the XY - plane we get a picture shown in Fig. 1.5.

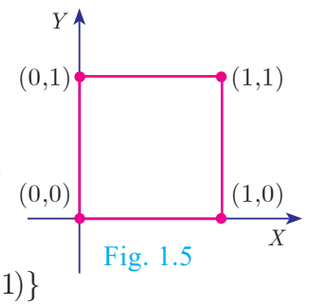


Fig. 1.5

$$(A\times B)\times C = \{(0,0),(0,1),(1,0),(1,1)\}\times\{0,1\}$$
$$= \{(0,0,0),(0,0,1),(0,1,0),(0,1,1),(1,0,0),(1,0,1)(1,1,0),(1,1,1)\}$$

Representing $A\times B\times C$ in the XYZ - space we get a picture as shown in Fig. 1.6.

Thus, $A\times B$ represent vertices of a square in two dimensions and $A\times B\times C$ represent vertices of a cube in three dimensions.

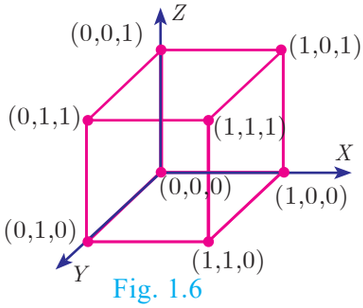


Fig. 1.6

Note

In general if we join the cartesian product of two non-empty sets provides a shape in two dimensions and similarly cartesian product of three non-empty sets provide an object in three dimensions.

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